

| Technology Stack                                  |                                                    |
|---------------------------------------------------|----------------------------------------------------|
| Requirement Analysis Phase — Architecture & Stack |                                                    |
| Project Name                                      | Automated Network Request Management in ServiceNow |
| Phase                                             | Phase 2: Requirement Analysis                      |
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| Platform / Domain                                 | ServiceNow / IT Service Management (ITSM)          |
| Academic Year                                     | 2025 – 2026                                        |
| 1. Technical Architecture Summary                 |                                                    |

The Automated Network Request Management solution utilizes ServiceNow's multi-layered enterprise cloud architecture. The system is organized into four primary layers, each responsible for a distinct aspect of the request lifecycle:

Presentation Layer

Provides structured form interfaces to end users through the unified ServiceNow Service Portal and native internal ITSM workspaces for engineers and approvers.

Application / Business Logic Layer

Enforces data security and validates complex network configuration inputs using Client Scripts and UI Policies. Manages role-based access through Access Control Lists (ACLs).

Automation Layer

Manages end-to-end request state routing using Flow Designer. Orchestrates multi-stage approval chains, conditional branching, SLA clock management, and event-driven email notification dispatch.

## Storage Layer

Securely houses standard ITSM tables (sc\_req\_item, sysapproval\_approver, sc\_task) and the custom u\_network\_database table in the ServiceNow cloud database with full audit trail support.

## 2. Components & Technologies

| S. No | Component           | Description                                                                                           | Technology                                                                                 |
|-------|---------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1     | User Interface      | Dedicated requester portal and centralized workspaces for network operators.                          | ServiceNow Service Portal, Service Catalog UI, Native Forms & Lists                        |
| 2     | Application Logic   | Dynamic conditional form behaviors, field hiding, regex parameter validation for IP/VLAN/port inputs. | UI Policies, Client Scripts, Data Policies                                                 |
| 3     | Workflow Automation | Multi-stage approval loops, sequential task distributions, and live operational escalations.          | Flow Designer, Core Approval Engine, Service Catalog Workflow                              |
| 4     | Notifications       | Automated event-driven email dispatch for all lifecycle stage transitions.                            | ServiceNow Notification Engine, Email Templates                                            |
| 5     | Data Storage        | Standard ITSM tables and custom u_network_database for operational logging.                           | ServiceNow Cloud Database (sc_req_item, sc_task, sysapproval_approver, u_network_database) |
| 6     | Security            | Role-based data access control and field-level permission enforcement.                                | Access Control Lists (ACLs), sys_user Roles                                                |
| 7     | SLA Management      | Real-time SLA tracking with automated escalation at configurable breach thresholds.                   | ServiceNow SLA Engine, Escalation Rules                                                    |
| 8     | Deployment          | Configuration packaging and environment migration without manual re-entry.                            | ServiceNow Update Sets                                                                     |
| 9     | Reporting           | Request volume monitoring, SLA performance dashboards, and productivity metrics.                      | ServiceNow Reporting & Performance Analytics                                               |

### 3. Technology Stack Justification

#### ServiceNow Platform

Chosen as the delivery platform due to its native ITSM modules, no-code/low-code automation capabilities, enterprise-grade security model, and proven scalability in large IT operations environments.

#### Flow Designer

Selected over legacy Workflow Editor due to its visual, action-based design interface, better support for conditional branching, and native integration with the SLA and notification engines.

#### Service Catalog

Used as the structured request intake mechanism because it provides consumer-grade usability for end users while enforcing backend data validation and workflow triggers automatically.

#### Client Scripts & UI Policies

Applied for real-time form-level validation and dynamic field visibility, ensuring data quality at the point of entry before submission rather than during back-end processing.